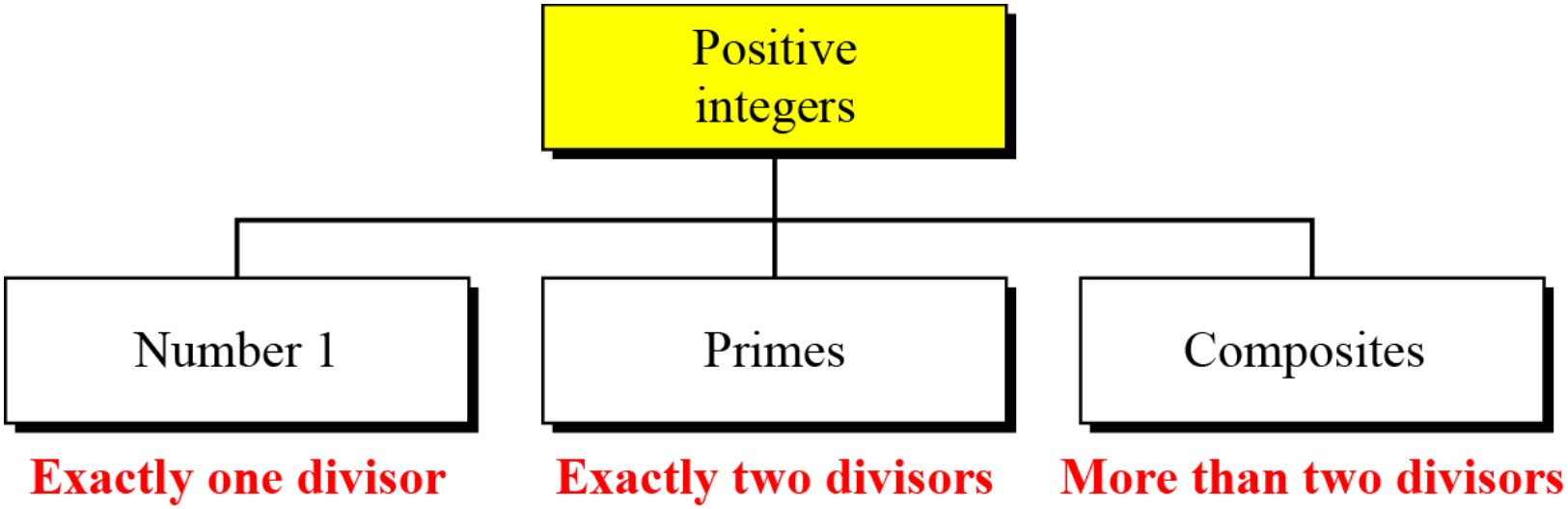
PRIMES

Asymmetric-key cryptography uses primes extensively. The topic of primes is a large part of any book on number theory. This section discusses only a few concepts and facts:

- **Fermat's Little Theorem**
- **Euler's Theorem**
- **Discrete Logarithms**

Definition: Prime

Three groups of positive integers



Note

A prime is divisible only by itself and 1.

Example

What is the smallest and largest primes?

Solution

The smallest prime is 2, which is divisible by 2 (itself) and 1.
The largest prime . . . ?

Example

List the primes smaller than 10.

Solution

There are four primes less than 10: 2, 3, 5, and 7. It is interesting to note that the percentage of primes in the range 1 to 10 is 40%. The percentage decreases as the range increases.

Cardinality of Primes

Infinite Number of Primes

Note

There is an infinite number of primes.

Checking for Primeness

Given a number n, how can we determine if n is a prime?

The answer is that we need to see if the number is divisible by all primes less than $\sqrt{n}$

This method is inefficient if the number is large, but it is a good start.

Example

Is 97 a prime?

Solution

The floor of $\sqrt{97} = 9$. The primes less than 9 are 2, 3, 5, and 7. We need to see if 97 is divisible by any of these numbers. It is not, so 97 is a prime.

Example

Is 301 a prime?

Solution

The floor of $\sqrt{301} = 17$. We need to check 2, 3, 5, 7, 11 and 13. The numbers 2, 3, and 5 do not divide 301, but 7 does. Therefore 301 is not a prime.

PRIMALITY TESTING

Finding an algorithm to correctly and efficiently test a very large integer and output a prime or a composite has always been a challenge in number theory, and consequently in cryptography. However, recent developments look very promising.

Topics discussed in this section:

Recommended Primality Test – ***Miller- Rabin Test***

Miller-Rabin Test

Miller-Rabin Test

$$n - 1 = m \times 2^k$$

Idea behind Fermat primality test

$$a^{n-1} = a^{m \times 2^k} = [a^m]^{2^k} = [a^m]^{2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}} \quad k \text{ times}$$

Note

The Miller-Rabin test needs from step 0 to step $k - 1$.

Continued

```
Miller_Rabin_Test (n, a)                    // n is the number; a is the base.
{
    Find m and k such that n − 1 = m × 2^k
    T ← a^m mod n
    if (T = ± 1)  return "a prime"
    for (i ← 1 to k − 1)                    // k − 1 is the maximum number of steps.
    {
        T ← T^2 mod n
        if (T = +1) return "a composite"
        if (T = −1) return "a prime"
    }
    return "a composite"
}
```

Continued

Example 9.25

Does the number 561 pass the Miller-Rabin test?

Solution

Using base 2, let $561 - 1 = 35 \times 2^4$, which means $m = 35$, $k = 4$, and $a = 2$.

Initialization:	$T = 2^{35} \bmod 561 = 263 \bmod 561$	
$k = 1$:	$T = 263^2 \bmod 561 = 166 \bmod 561$	
$k = 2$:	$T = 166^2 \bmod 561 = 67 \bmod 561$	
$k = 3$:	$T = 67^2 \bmod 561 = +1 \bmod 561$	$\rightarrow$ **a composite**

Continued

Example 9.26

We already know that 27 is not a prime. Let us apply the Miller-Rabin test.

Solution

With base 2, let $27 - 1 = 13 \times 2^1$,

which means that $m = 13$, $k = 1$, and $a = 2$.

In this case, because $k - 1 = 0$,

we should do only the initialization step:

$$T = 2^{13} \bmod 27 = 11 \bmod 27.$$

However, because the algorithm never enters the loop, it returns a composite.

Continued

Example 9.27

We know that 61 is a prime, let us see if it passes the Miller-Rabin test.

Solution

We use base 2.

$$61 - 1 = 15 \times 2^2 \rightarrow \quad m = 15 \quad k = 2 \quad a = 2$$

$$\textit{Initialization:} \quad \mathrm{T} = 2^{15} \bmod 61 = 11 \bmod 61$$

$$k = 1 \qquad \mathrm{T} = 11^2 \bmod 61 = -1 \bmod 61 \qquad \rightarrow \textbf{a prime}$$

Example

Finally we test an integer that is in fact prime, $n = 104513$, using $a = 3$.

$$n - 1 = 2^6 \cdot 1633.$$

$$\begin{aligned}
3^{1633} &\equiv 88958 && \pmod{n} \\
3^{2\cdot 1633} &\equiv 88958^2 \equiv 10430 && \pmod{n} \\
3^{2^2\cdot 1633} &\equiv 10430^2 \equiv 91380 && \pmod{n} \\
3^{2^3\cdot 1633} &\equiv 91380^2 \equiv 29239 && \pmod{n} \\
3^{2^4\cdot 1633} &\equiv 29239^2 \equiv 2781 && \pmod{n} \\
3^{2^5\cdot 1633} &\equiv 2781^2 \equiv -1 && \pmod{n}
\end{aligned}$$

The test concludes that *n* is a probable prime.

Example – The Rabin Miller Test

$\boldsymbol{n} = \mathbf{252601}$, $n-1 = 2^3 \cdot 31575$. Choose $a = 85132$. $a^{31575} \equiv 191102 \pmod{n}$ $a^{2\cdot 31575} \equiv 184829 \pmod{n}$ $a^{2^2\cdot 31575} \equiv 1 \pmod{n}$ *Conclusion:* n is ***composite***. (184829 is a square root of 1, mod n, different from ± 1.)	$\boldsymbol{n} = \mathbf{3057601}$, $n-1 = 2^6 \cdot 47775$. Choose $a = 99908 \pmod{n}$. $a^{47775} \equiv 1193206 \pmod{n}$ $a^{2\cdot 47775} \equiv 2286397 \pmod{n}$ $a^{2^2\cdot 47775} \equiv 235899 \pmod{n}$ $a^{2^3\cdot 47775} \equiv 1 \pmod{n}$ *Conclusion:* n is ***composite***. (235899 is a square root of 1, mod n, different from ± 1.)

Example

$\boldsymbol{n} = \mathbf{101089}$, $n-1 = 2^5 \cdot 3159$. Choose $a = 5$. $a^{3159} \equiv 101088 \equiv -1 \pmod{n}$ *Conclusion:* n is ***probably prime***.	$\boldsymbol{n} = \mathbf{280001}$, $n-1 = 2^6 \cdot 4375$. Choose $a = 105532$. $a^{4375} \equiv 236926 \pmod{n}$ $a^{2 \cdot 4375} \equiv 168999 \pmod{n}$ $a^{2^2 \cdot 4375} \equiv 280000 \equiv -1 \pmod{n}$ *Conclusion:* n is ***probably prime***.

Continued

Example 9.28

The number 4033 is a composite (37 × 109). Does it pass the recommended primality test?

Solution

1. **Perform the divisibility tests first. The numbers 2, 3, 5, 7, 11, 17, and 23 are not divisors of 4033.**
2. **Perform the Miller-Rabin test with a base of 2,**

$$4033 - 1 = 63 \times 2^6, \text{ which means } m \text{ is 63 and } k \text{ is 6.}$$

Initialization: $T \equiv 2^{63} \pmod{4033} \equiv 3521 \pmod{4033}$

$k = 1$ $\quad T \equiv T^2 \equiv 3521^2 \pmod{4033} \equiv -1 \pmod{4033}$ $\rightarrow$ **Passes**

Euler's Phi-Function

Euler's phi-function, ϕ (n) counts the positive integers up to a given integer n that are relatively prime to n.

which is sometimes called the Euler's totient function plays a very important role in cryptography.

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1. $\phi(1) = 0.$
2. $\phi(p) = p - 1$ if p is a prime.
3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m and n are relatively prime.
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Continued . . .

We can combine the above four rules to find the value of $\phi(n)$. For example, if n can be factored as

$$n = p_1^{e_1} \times p_2^{e_2} \times \ldots \times p_k^{e_k}$$

then we combine the third and the fourth rule to find

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) \times (p_2^{e_2} - p_2^{e_2-1}) \times \cdots \times (p_k^{e_k} - p_k^{e_k-1})$$

Note

The difficulty of finding $\phi(n)$ depends on the difficulty of finding the factorization of *n*.

Continued

Example

What is the value of $\phi(13)$?

Solution

Because 13 is a prime, $\phi(13) = (13 - 1) = 12$.

Example

What is the value of $\phi(10)$?

Solution

We can use the third rule:

$$\phi(10) = \phi(2) \times \phi(5) = (2-1) \times (5-1)$$

$$= 1 \times 4 = 4, \qquad \text{because 2 and 5 are primes.}$$

Continued

Example

What is the value of $\phi(240)$?

Solution

We can write $240 = 2^4 \times 3^1 \times 5^1$. Then

$$\phi(240) = (2^4 - 2^3) \times (3^1 - 3^0) \times (5^1 - 5^0) = 64$$

Example

Can we say that $\phi(49) = \phi(7) \times \phi(7) = 6 \times 6 = 36$?

Solution

No. The third rule applies when *m* and *n* are relatively prime. Here $49 = 7^2$. We need to use the fourth rule: $\phi(49) = 7^2 - 7^1 = 42$.

Continued

Example

What is the number of elements in Z_{14}*?

Solution

The answer is $\phi(14) = \phi(7) \times \phi(2) = 6 \times 1 = 6$. The members are 1, 3, 5, 9, 11, and 13.

Note

Interesting point: If $n > 2$, the value of $\phi(n)$ is even.

Fermat's Little Theorem

`p` is a prime and `gcd(a,p)=1`

First Version

$$\boxed{a^{p-1} \equiv 1 \bmod p}$$

Second Version

$$\boxed{a^{p} \equiv a \bmod p}$$

Continued . . .

Example

Find the result of 6^{10} mod 11.

Solution

We have 6^{10} mod 11 = 1. This is the first version of Fermat's little theorem where p = 11.

Example

Find the result of 3^{12} mod 11.

Solution

Here the exponent (12) and the modulus (11) are not the same. With substitution this can be solved using Fermat's little theorem.

$$3^{12} \bmod 11 = (3^{11} \times 3) \bmod 11 = (3^{11} \bmod 11)\,(3 \bmod 11) = (3 \times 3) \bmod 11 = 9$$

Euler's Theorem

First Version

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Second Version

$$a^{k \times \phi(n) + 1} \equiv a \pmod{n}$$

- where `gcd(a,n)=1`

Note

The second version of Euler's theorem is used in the RSA cryptosystem.

Example:

1. $a=3; n=10; \quad \varnothing(10)=4;$

hence $3^4 = 81 = 1 \bmod 10$

2. $a=2; n=11; \varnothing(11)=10;$

hence $2^{10} = 1024 = 1 \bmod 11$

Continued

Example

Find the result of 6^{24} mod 35.

Solution

We have $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35 = 1$.

Example

Find the result of 20^{62} mod 77.

Solution

If we let $k = 1$ on the second version, we have $\varnothing(77)=60$

$$20^{62} \bmod 77 = (20 \bmod 77)\,(20^{\phi(77)+1} \bmod 77) \bmod 77$$
$$= (20)(20) \bmod 77 = 15.$$

Continued

Example

Find the result of 2^{43210} mod 101.

Solution

Note that 101 is prime. From Euler's theorem, we know that

$$2^{100} = 1 \bmod 101.$$

Therefore,

$$2^{43210} = (2^{100 \times 432})\, 2^{10} = 1^{432}\, 2^{10} = 1024 = 14 \pmod{101}$$

Practice Problems

1) Use the CRT to find x for the following equations:

a) $x \equiv 3 \pmod 5$, $x \equiv 1 \pmod 7$, $x \equiv 6 \pmod 8$

b) $x \equiv 1 \pmod 3$, $x \equiv 4 \pmod 5$, $x \equiv 6 \pmod 7$

c) $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$

d) $x \equiv 3 \pmod 7$, $x \equiv 3 \pmod{13}$, $x \equiv 0 \pmod{12}$

2) Using Fermat's Theorem, find $3^{201} \bmod 11$.

3) Compute the following: a) $\phi(27)$ b) $\phi(41)$ c) $\phi(440)$ d) $\phi(231)$